

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Rising Waters codebase in terms of structure, readability, maintainability, and reusability. The project follows a modular approach using Python, Flask, and Machine Learning libraries. Proper naming conventions, folder organization, and reusable functions improve code maintainability and future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs use throughout the code	Yes	Proper indentation maintained throughout the project
2	Proper File Structure	Files and folders are logically organized	Yes	Files are organized systematically
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variable names clearly indicate their purpose
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named appropriately
5	Code Comments	Inline and block comments explain logic	Partial	Important sections are commented
6	Modular Design	Code is split into reusable functions/modules	yes	Code is divided into reusable modules
7	No Redundant Code	Duplicate or unused code is removed	yes	Unnecessary code has been removed
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation is provided

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model training and prediction	High

2	Prediction Module	Scikit-Learn	Flask application	High
3	Flask Routes	Flask	Web request handling	High
4	HTML Templates	HTML/CSS	User interface pages	Medium
5	Input Validation Functions	Python	User input processing	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized and modular structure
Readability	4	Easy to understand and maintain
Reusability	5	Modules can be reused effectively
Documentation / Comments	4	Sufficient comments for understanding
Overall Score	4.5/5	Good quality and maintainable code